Identifying persistent structures in multiscale ‘omics data

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Abstract

In any ‘omics study, significant patterns in data often become apparent only when looking at the right scale. For instance, when clustering single-cell transcriptomes, is the analysis tuned to discover broad or specific cell types? Likewise, protein communities revealed from protein networks can vary widely in sizes depending on the method. Here, we use the concept of “persistent homology” drawn from mathematical topology, to develop a computational method, termed HiDeF (Hierarchical community Decoding Framework) for identifying robust structures in data at all scales simultaneously. HiDeF outperformed alternative approaches that do not consider the persistence of structures in benchmark experiments with simulated data and real-world datasets of single-cell transcriptomics and protein interactions. Comparative analysis of protein-protein interaction networks found significant differences in the distributions of community sizes across networks generated with different measurement approaches. HiDeF could facilitate biological discovery within large ‘omics data, such as expanding the catalog of identified cell types in single-cell transcriptomics data and identifying protein modules from multi-omics protein networks. The method has been integrated into CDAPS (Community Detection APplication and Service), a Cytoscape-based software framework, and also available as a Python package.

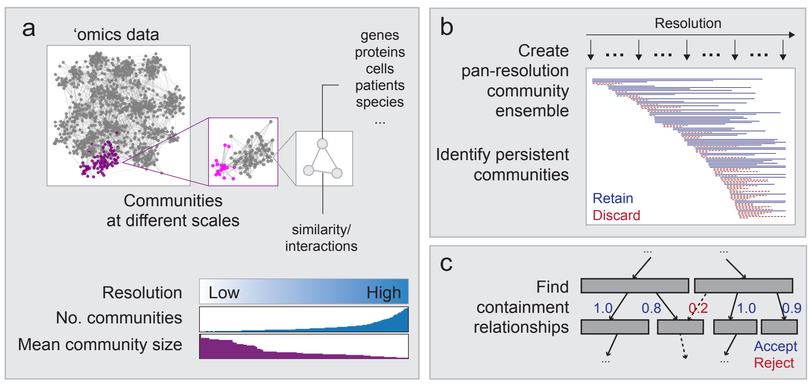

Fig. 1. Identification of persistent communities with HiDeF. (a) ‘Omics data often contain community structures at different spatial resolutions. (b) Pan-resolution community detection yields a candidate pool of communities. Persistent communities that are robustly identified across a wide range of resolutions. (c) Set containment analysis defines the hierarchical relationships between communities.

Single-cell clustering

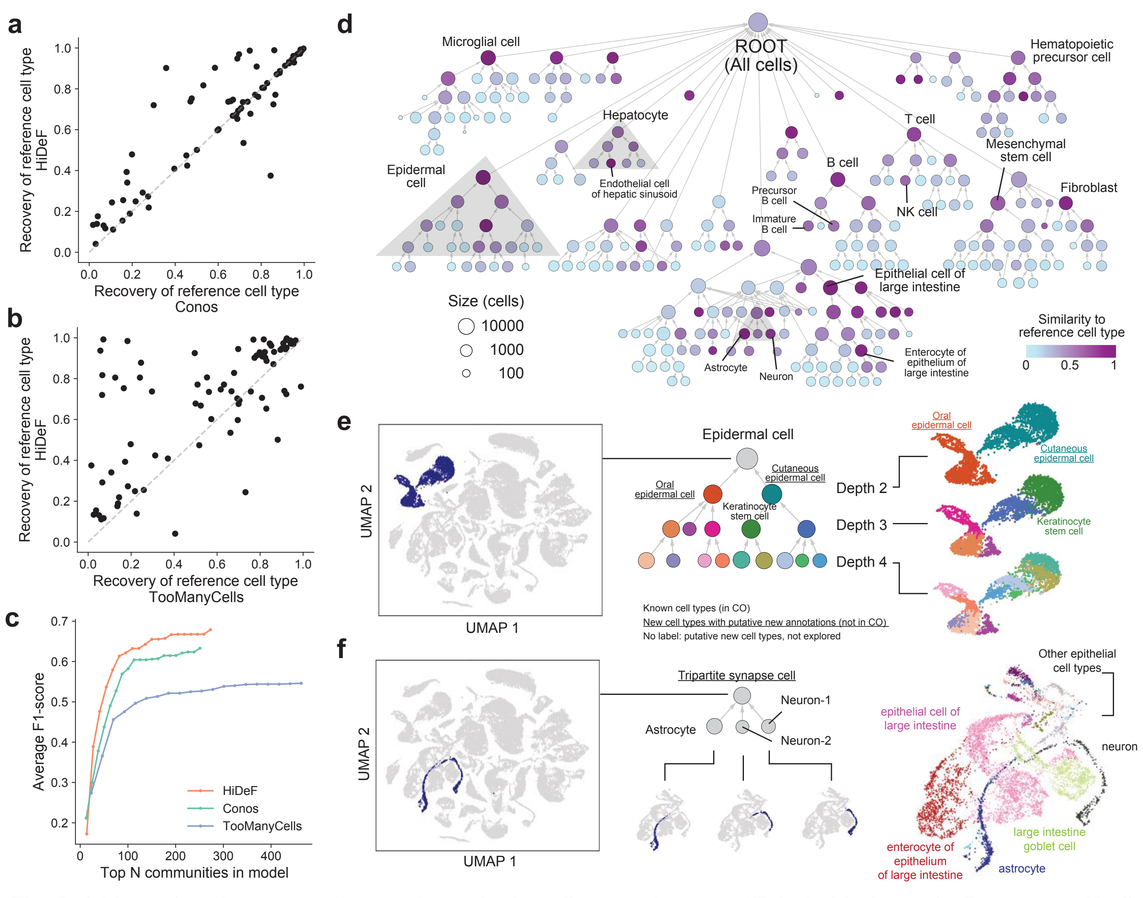

Fig. 2. A hierarchy of mouse cell types from single-cell transcriptomes (*Tabula Muris*). (a-b) Recovery of individual reference cell types by HiDeF (y axis) in comparison to alternative tools (x-axis). (c) Recovery of reference cell types among the top N ranked cell communities. (d) Hierarchy of putative cell types identified by HiDeF. Color gradients on vertices indicating the extent of the optimal match to a reference type. (e-f) HiDeF captured cell populations uncharacterized in the original analysis, e.g. a cluster joining epidermal cells from skin and tongue, and a cluster joining astrocytes and neuron cells that is difficult to distinguish via 2D projection.

Protein module analysis

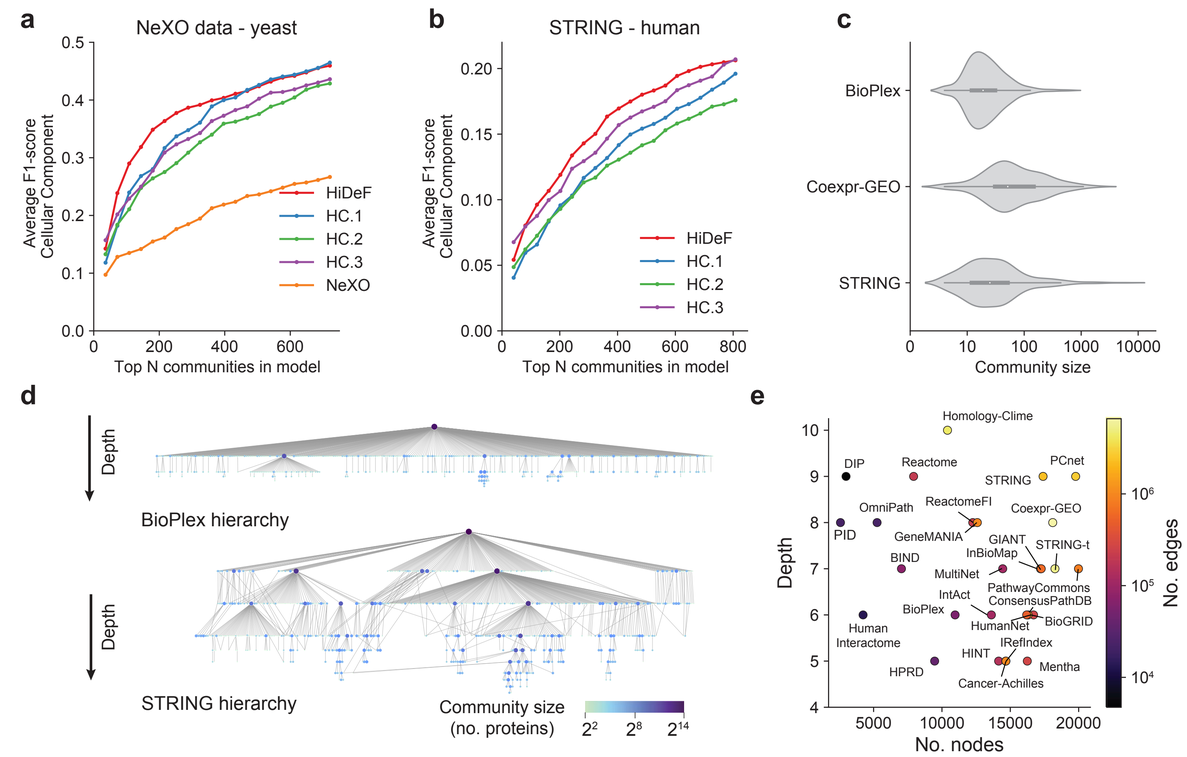

Fig. 3. Hierarchical structure of protein networks. (a-b) Recovery of cellular components in Gene Ontology by community detection methods (colored traces) versus number of top communities examined. HC.1-3 represent hierarchical clustering of pairwise distances generated by network embedding tools (Mashup, DSD, and deepNF). (c) Distributions of community sizes for three human protein networks of different data types. (d) Hierarchical structures identified for BioPlex 2.0 or STRING databases. (e) 27 public databases of protein-protein interaction networks were analyzed by HiDeF and profiled by the maximum depths of their resulting hierarchies.

Cytoscape integration

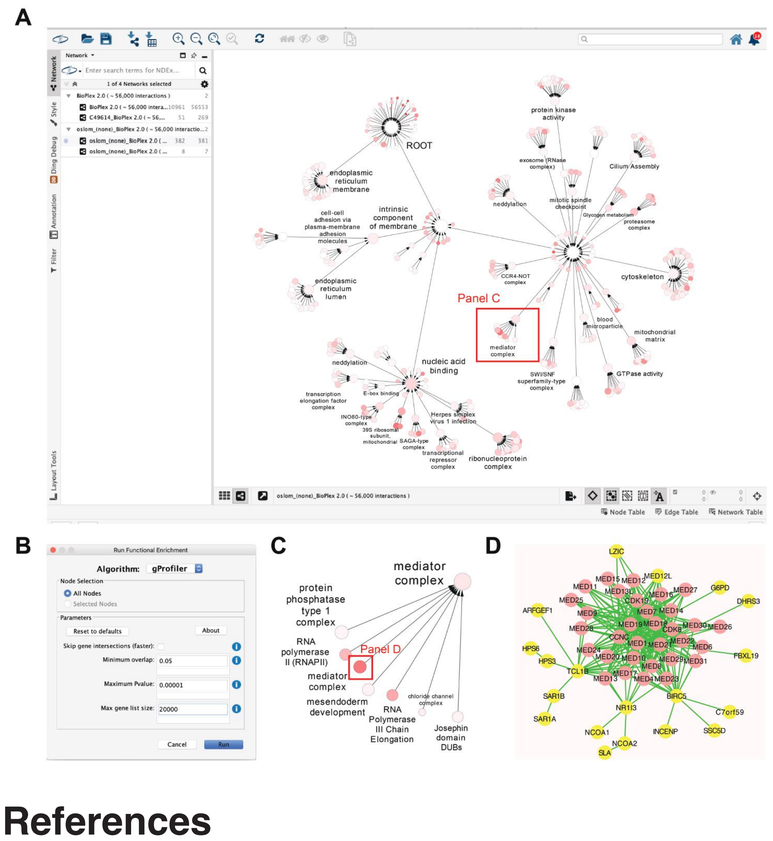

Fig. 4. HiDeF has been integrated into the CDAPS (Community Detection APplication and Service) software framework in Cytoscape. (a) A hierarchy network generated by multiscale community detection. (b-c) Protein communities can be annotated by functional enrichment services. (d) The interaction network supporting a community can be easily generated and visualized, and compared to gene sets in existing databases (node colors).

Access HiDeF

Access CDAPS

References

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Singhal, A., Cao, S., Churas, C., Pratt, D., Fortunato, S., Zheng, F., & Ideker, T. (2020). Multiscale community detection in Cytoscape. PLoS computational biology, 16(10), e1008239.